

Lecture 13

ChE 467/596-012 Polymer Rheology

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Exam Problem #1

- **Problem 1** (18) *True or False* (you can provide justification if you want)
- Rod climbing effect observed for polymeric liquids is often attributed to normal stresses.
- In shear flow, the distance between neighboring particles changes very slowly with time.
- Extensional flows are sometimes called strong flows whereas shear flow are often referred to as common flow.
- Die swelling during polymer extrusion decreases with increasing L/D ratio of the die.
- Dynamic experiments have to be done at very small strain amplitude to avoid equipment damage.
- The plateau modulus of a polymer is independent of molecular weight and temperature.

Exam Problem #2

Problem 2 (14)

Apple sauce at 30°C may be described by the following constitutive equation: $\tau_{yx} = 6.0\dot{\gamma}^{0.5}$

Honey at 30°C obeys the following model: $\tau_{yx} = 9.0\dot{\gamma}$

- (a) (8) Evaluate the viscosities for apple sauce and honey at shear rates of 1, and 9 sec^{-1} .
- (b) (6) Name the constitutive models that describe the two substances and state the general differences (as many as possible) between them.

(a) Apple sauce viscosities: 6 and 2; Honey viscosity 9.

(b) Honey is Newtonian; apple sauce is power law.

Exam Problem #3

Problem 3 (10)

(a) (4) Consider the vector/tensor operation given by: $\underline{v} \cdot \underline{w} \cdot \underline{z} \underline{p}$
[Note there is no operator between vectors $\underline{z}$ and $\underline{p}$] Does this yield a tensor, vector or scalar?

(b) (6) Write down the end product of this operation in terms of index notation. NOTE: Do not expand!! Keep your final answer in terms of i, j, k (e.g., $v_i w_j z_j \underline{e}_i$ or whatever)

(a) scalar

(b) $v_i w_j z_j p_i$, in other words v and p have same index; w and z have same index. U have to take dot product of w and z first; and then between v and p .

Exam Problem #4

Problem 4 (15)

The velocity field for a material has the general form: $\underline{v} = [v_x, v_y, 0]$; i.e., has v_x and v_y component.

Here $v_y = U(1 - bx^2)$, $v_x = U(1 + 2bxy)$ with U and b being constants

- (a) Given that the fluid is Newtonian, what type of a flow is it?
- (b) If the fluid were Non-Newtonian, what type of flow(s) could this be describing?
- (c) Is the flow homogeneous?

Exam Problem#5

Problem 5 (12)

A solution is being characterized in a cone and plate rheometer, at a specified temperature in a steady shear experiment, and the cone angle is 0.01 radians. At a rotational speed of 1 radian/sec, the normal force exerted by the liquid is 10^5 dynes, and the torque signal measured is $10^6 \text{ g cm}^2 / \text{sec}^2$

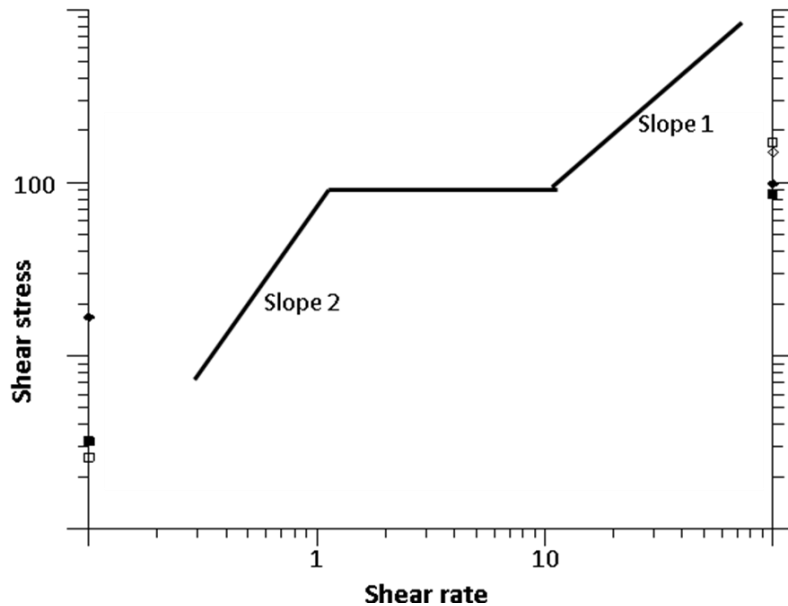
(a) (7) Estimate the shear rate and viscosity of the material for the specified condition. The plate diameter is 5 cm.

(b) (5) What would you classify (e.g., Newtonian, viscoelastic, Bingham, etc, etc) this solution as, and why?

$$\tau = \frac{3 \times \text{torque}}{2\pi R^3} = 30557.75$$

$$\dot{\gamma} = \frac{\text{rotation rate}}{\text{cone angle}} = 100$$

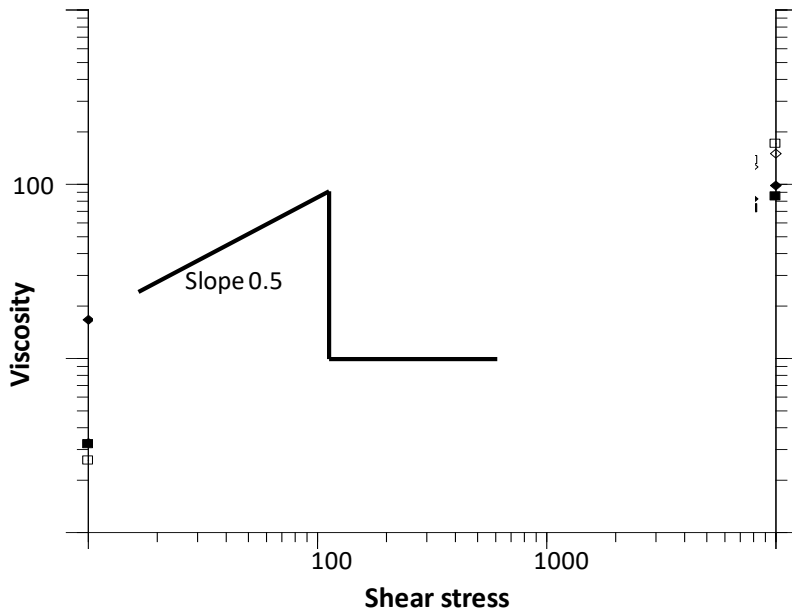
Solution-Exam Problem#6



Plot viscosity versus shear stress from the given information. Please sketch with as much details as possible (e.g., slope, numbers wherever possible, etc).

Solution-Exam Problem#6

Plot viscosity versus shear stress from the given information. Please sketch with as much details as possible (e.g., slope, numbers wherever possible, etc).



Solution-Exam Problem#7

Problem 7 (10)

Your friend from the Macromolecular Science Department suggests that non-linear properties (such as the full viscosity profile showing the shear thinning part) can never be obtained from linear measurements. You think otherwise. Suggest two experiments (linear) that you could do to obtain such non-linear properties (e.g., entire steady viscosity profile).

Solution-Exam Problem#8

Problem 8 (6)

Your friend has carefully calculated the stress tensor for a complex flow and come up with the following total stress tensor.

$$\underline{\underline{T}} = \begin{bmatrix} 20 & -8 & 0 \\ 8 & 15 & -3 \\ 0 & 3 & 15 \end{bmatrix}$$

Comment on the stress tensor and the nature of the flow it suggests (assume Newtonian fluid if you have to).

The stress is not symmetric and is not therefore a valid stress tensor, and so does not represent any flow.

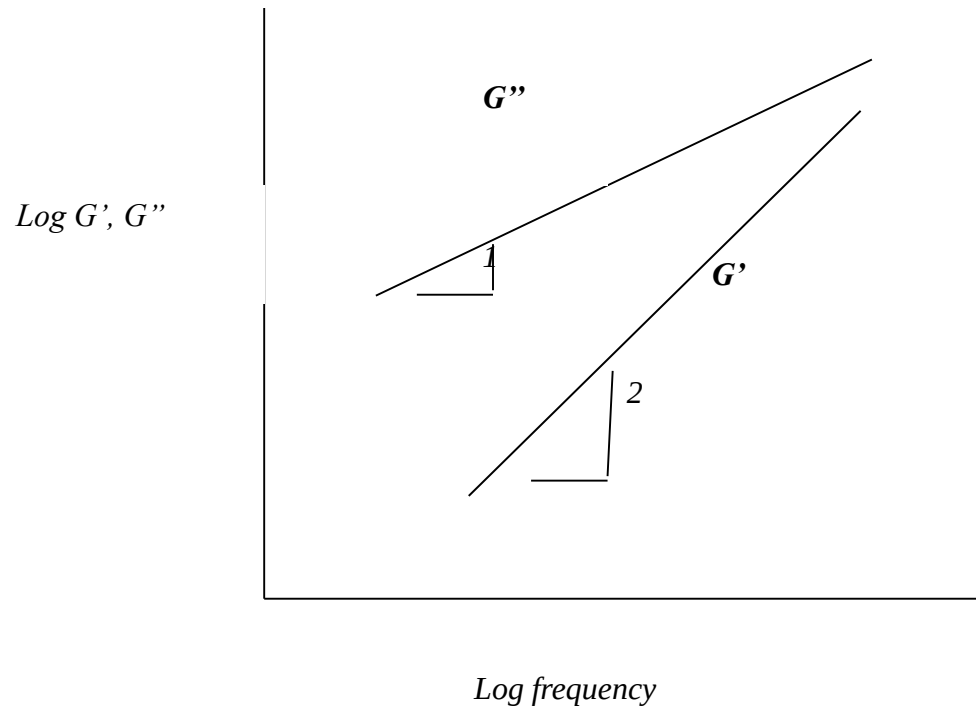
#4. (a) (4) Consider the vector/tensor operation given by: $\underline{\underline{\mathbf{A}}} \bullet \underline{\underline{\mathbf{B}}} : \underline{\underline{\mathbf{B}}} \bullet \underline{\underline{\mathbf{A}}}$. [Here, $\underline{\underline{\mathbf{A}}}$ and $\underline{\underline{\mathbf{B}}}$ are both tensors]. Does this yield a tensor, vector or scalar? **DO NOT SOLVE.**

(b) (6) Describe two flow situations (e.g. pulling a polymer film at an exponential rate) in which the total stress tensor, $\underline{\underline{\mathbf{T}}}$, has the following non-zero components.

$$T_{12} = T_{21} = \text{non-zero}$$

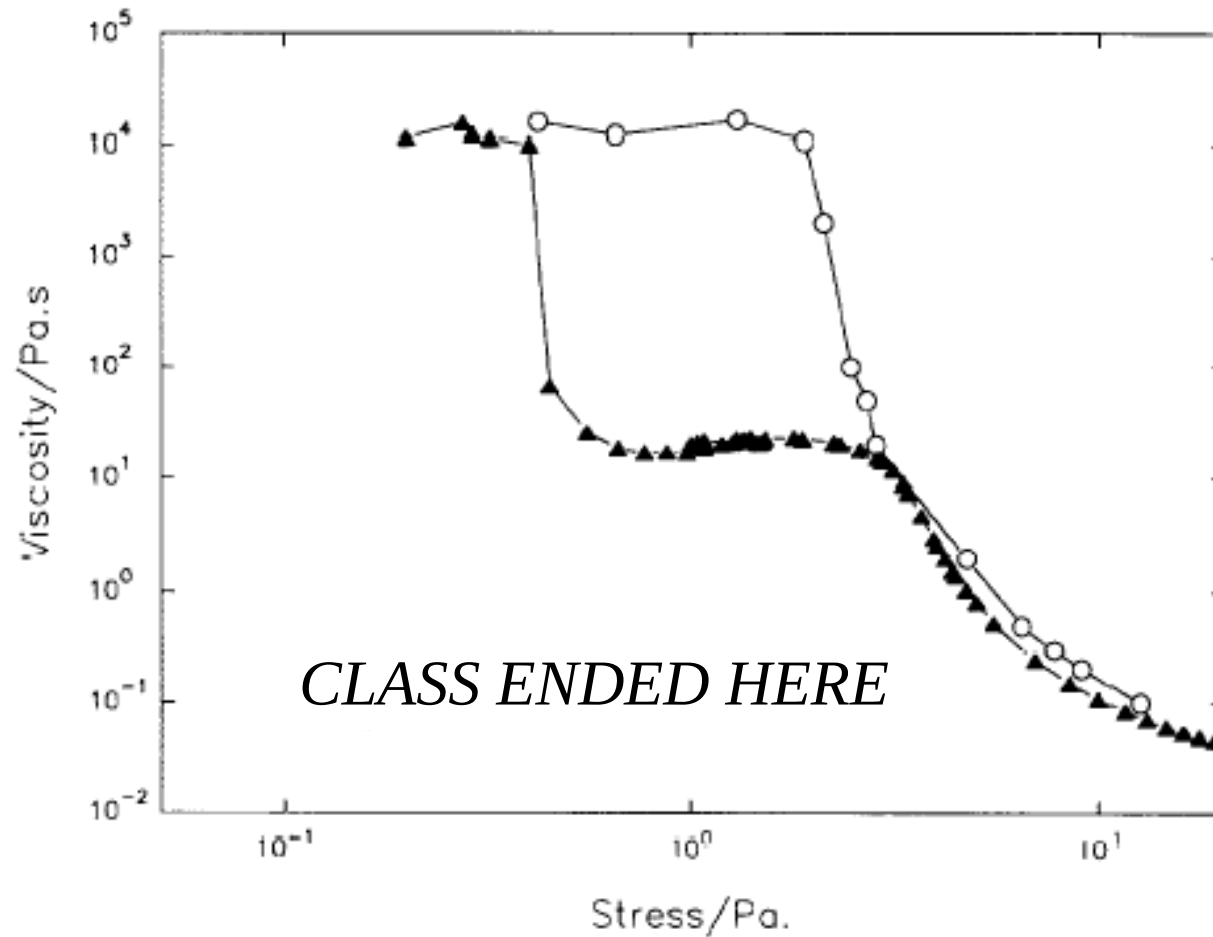
$$T_{11} - T_{22} = N_1 = \text{non-zero}; T_{22} - T_{33} = N_2 = \text{no-zero}$$

All other stress components are zero.



For this frequency range, please plot the complex viscosity as a function of frequency on a log-log scale.



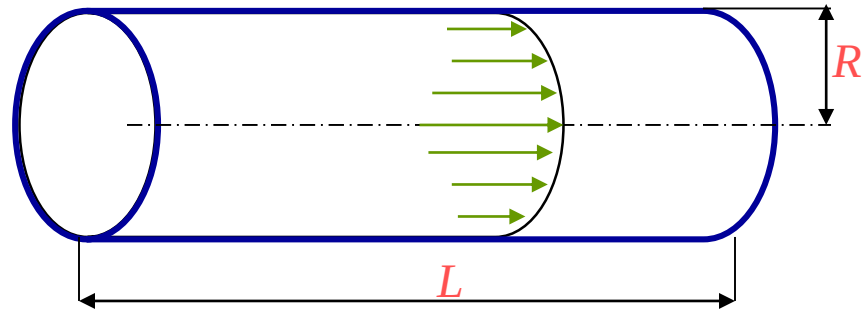


viscosity vs shear stress data for the same sample. In one case the data has been obtained using a vane geometry; in another case, the data has been obtained using a parallel plate geometry with smooth plates. Explain what you think is (are) the reasons for the discrepancy in the data. Also explain which data correspond to which geometry.

Pressure Driven Flow (Capillary Rheometer)

Principle => pipe flow

Non-homogenous flow



Measure shear stress at wall:

$$\tau_{\text{wall}} = \tau_{12} \Big|_{r=R} = \frac{R}{2} \frac{\Delta p}{L}$$

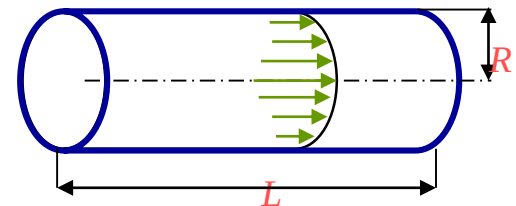
Measure flowrate:

$$\dot{\gamma}_{\text{wall}} = \frac{Q}{\pi R^3} \left[3 + \frac{d \ln Q}{d \ln \tau_{\text{wall}}} \right]$$

$$\eta = \frac{\tau_{\text{wall}}}{\dot{\gamma}_{\text{wall}}}$$

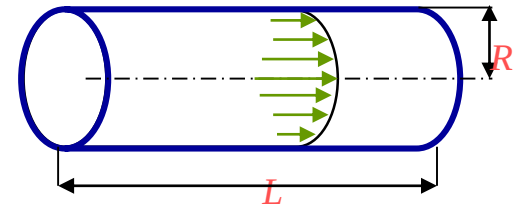
Capillary Rheometers-Disadvantages

- **Non-homogenous flow** (*Correction factor needed*)
- **High shear rates only**
- **Difficult to Clean** (*sample sticks on wall, solidification*)
- **Can Only measure viscosity**
 - *No normal stresses*
 - *No dynamic properties (G' , G'' , etc)*
- **Compressibility issues**
- **Viscous heating**
- **Pressure effect on η**



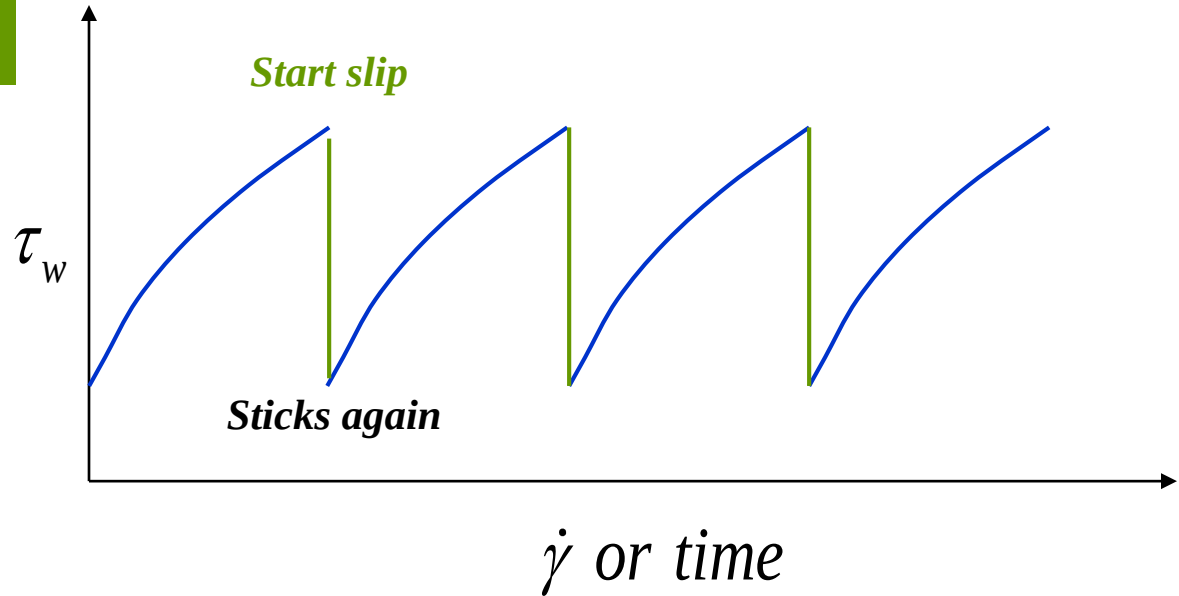
Capillary Rheometer-Advantages

- **No sample evaporation** (*Temperature studies*)
- **Can get high shear rate data**
 - *Relevant in processing*
 - *Expands the shear rate regime*
- **Similar to process flow** (*can have it as online device*)
- **Enclosed** (*good for volatile materials*)
- **Can use high pressure**



Capillary rheometry –other issues

Stick Slip



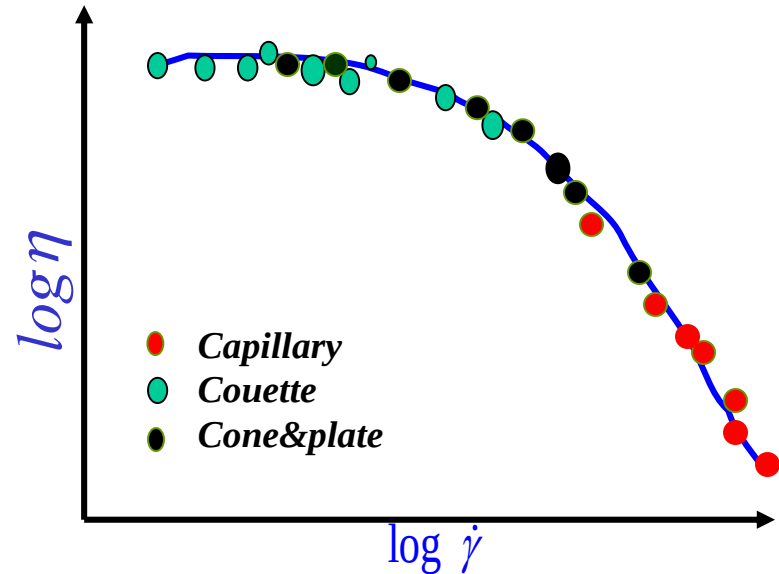
Melt

High shear rates $\rightarrow$ unstable flow $\rightarrow$ extrudate appears distorted

Jagged polymer samples coming out

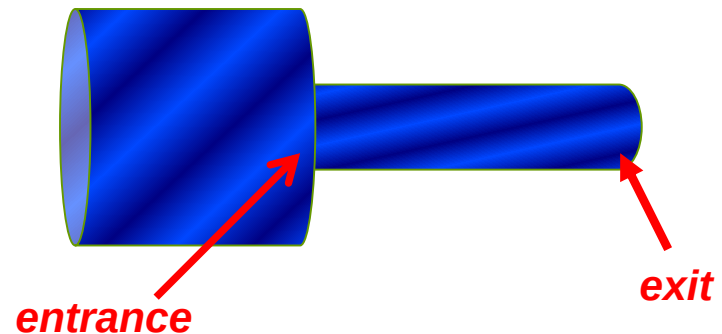
Capillary Rheometry-Other Issues

•Data can be combined with data from couette and cone & plate rheometers to get wide shear rate range



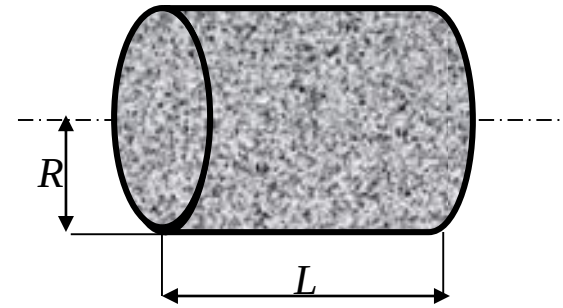
Wall slip – need to correct shear rate

Entrance/exit effects
-need to correct stress



Wall Slip Effects

- *Wall slip can happen, particularly if you have filled systems*
- *Fillers migrate inwards → particle lean layer near wall with much lower viscosity → slip layer*



To check for wall slip:

Measure viscosity at similar L/R ratio but different R

If results overlap => no wall slip

If different results => wall slip present

Wall Slip Effects, cont'd

Correcting shear rate for wall slip

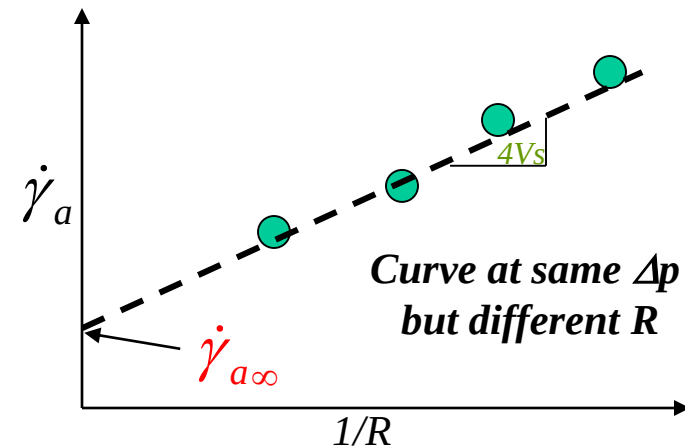
- Pick capillaries of different radii R
- Run experiment with each tube keeping Δp (i.e. τ_{wall}) same
- Plot app. shear rate vs $1/R$
- $\dot{\gamma}_{a\infty}$ is the corrected shear rate

$$\dot{\gamma}_a = \dot{\gamma}_{a\infty} + \frac{4V_s}{R}$$

Get viscosity from $\eta = \frac{\tau_{\text{wall}}}{\dot{\gamma}_{a\infty}}$

$$\dot{\gamma}_a = \frac{Q}{\pi R^3} \left[3 + \frac{d \ln Q}{d \ln \tau_{\text{wall}}} \right]$$

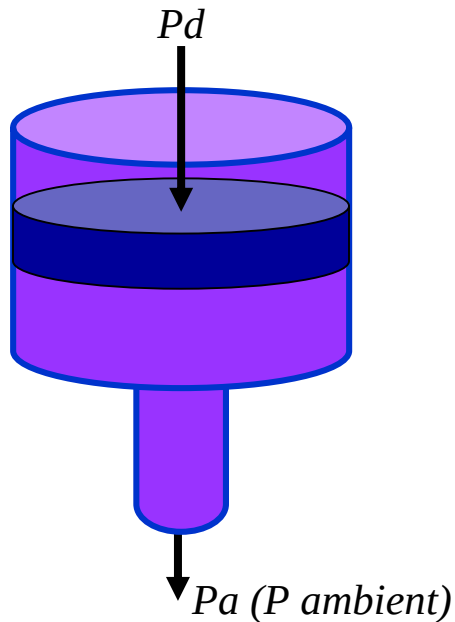
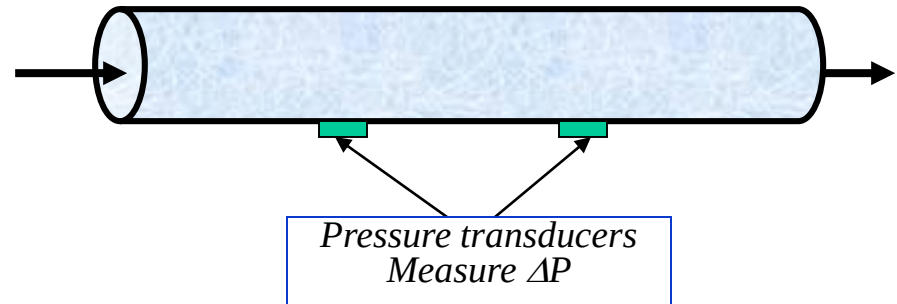
$\dot{\gamma}_a$ = apparent shear rate



- Repeat with different Δp to get different corrected shear rate

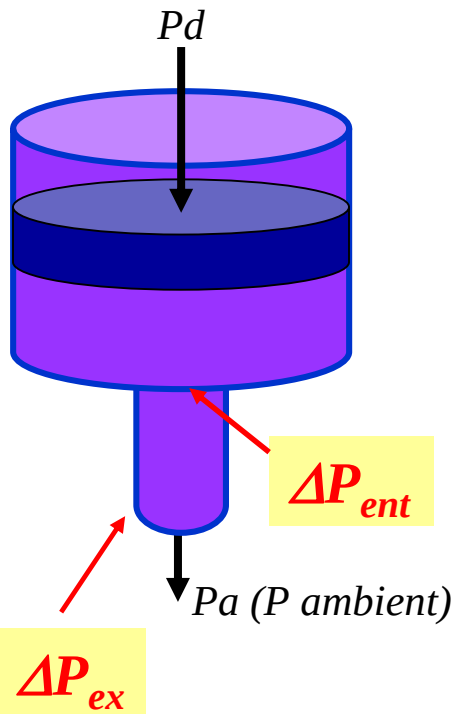
End Pressure Effects

1. Place pressure transducers far from the entrance and exit
No end pressure effect



2. Drive piston at pressure P_d ; material comes out at ambient pressure P_a
Need end pressure correction

End Pressure Effects



Measure $P_d - P_a$

$$\text{But } P_d - P_a = \Delta P_{ent} + \Delta P + \Delta P_{ex}$$

ΔP_{ent}
xs pressure from
entrance flow

ΔP
pressure from flow

ΔP_{ex}
xs pressure from
exit flow

$\Delta P_{end} = \Delta P_{ent} + \Delta P_{ex}$ introduces errors and over predicts stress if not taken into account. How?

Because measured $P_d - P_a$ often mistakenly taken as $\Delta P = 2\tau_w L/R$

End Pressure Effects-Bagley Correction

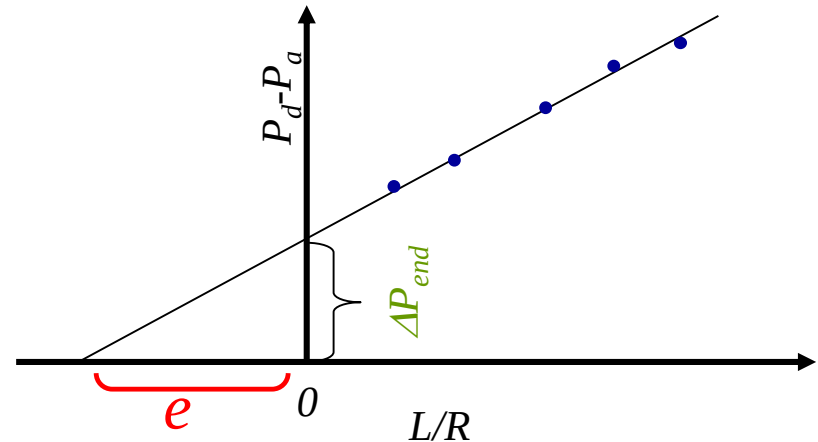
To correct for end pressure effect:

- *Keep Q constant and measure P_d versus R (P_a known)*
- *Plot $P_d - P_a$ vs L/R*
- *From Q get shear rate*
- *Get τ from the equation*

$$P_d - P_a = \frac{2\tau_w L}{R} + \Delta P_{end}$$

OR

$$\tau_w = \frac{P_d - P_a}{2(L/R + e)}$$

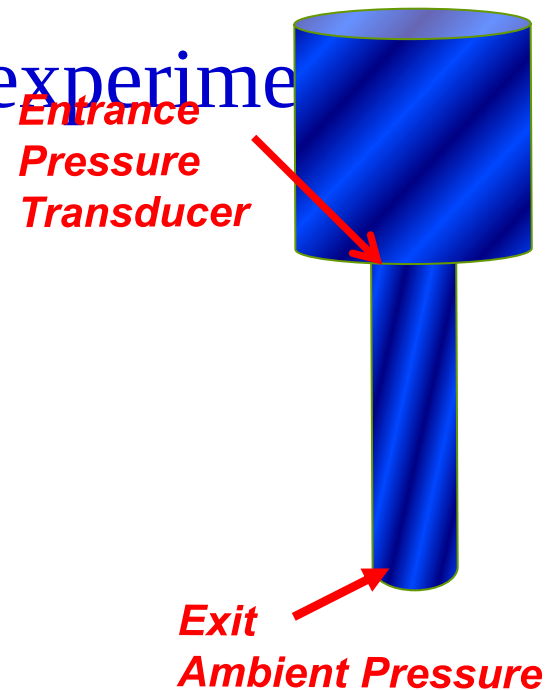


Data analysis

- AMS's lab in Venezuela 1999.
- Capillary pressure driven experiment
- Polypropylene at 200°C.
 - *What do we correct for?*

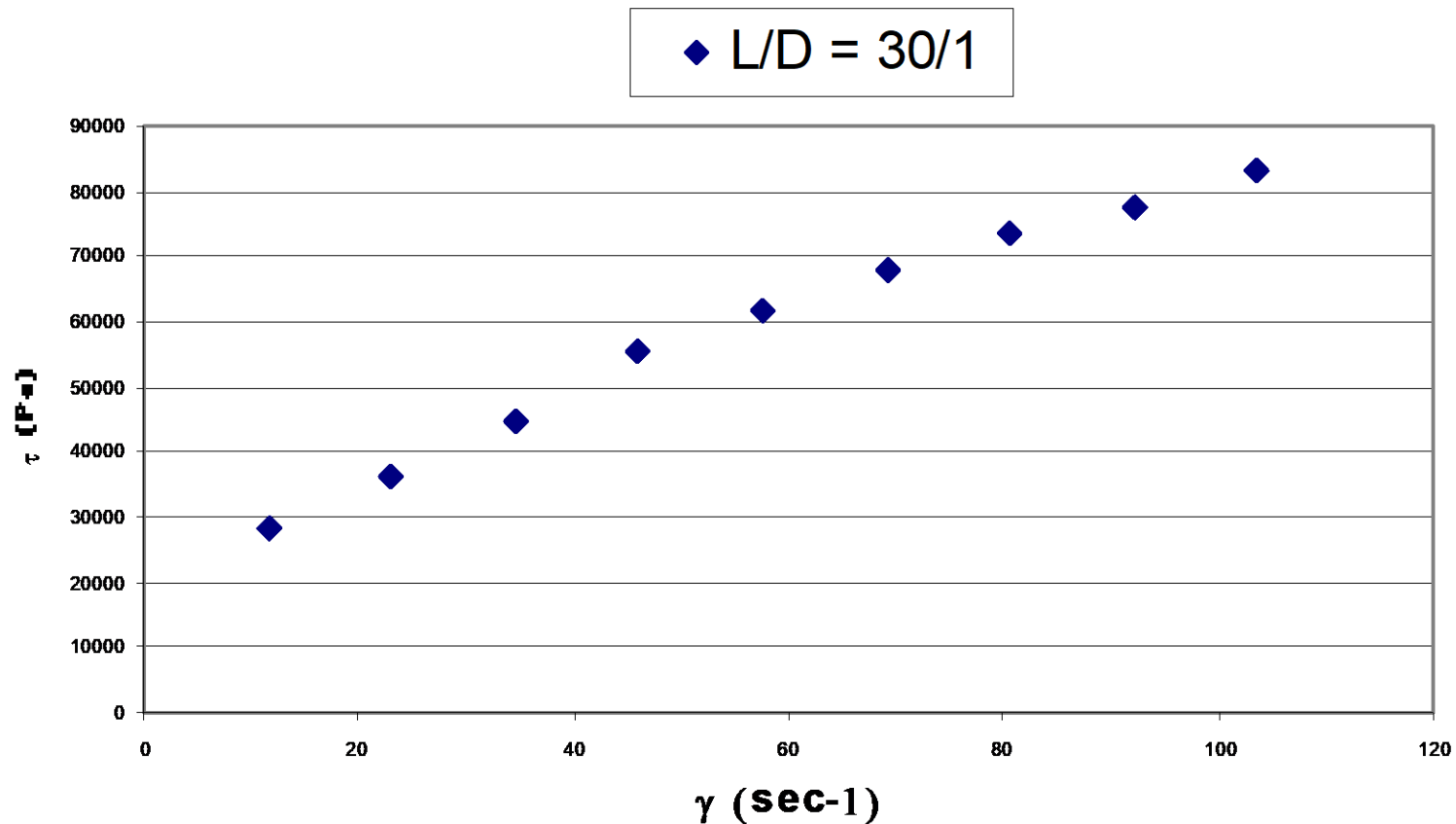
- *End effects*

- *Non-homogeneous flow*



Shear Flow Rheometry Experiment

Original Data: Position of Corrected Data?



Shear Flow Rheometry Experiment

- L/D ratios used:
 - 10mm/1mm
 - 20mm/1mm
 - 30mm/1 mm
- Corrections needed for stress τ_w and shear rate $\dot{\gamma}$.
 - Bagley plot:

$$\tau_w = \frac{P_d - P_a}{2(L/R + e)}$$

- Weissenberg-Rabinowitsch correction:

(For power law model)

$$\dot{\gamma}_{wall} = \frac{Q}{\pi R^3} \left[3 + \frac{d \ln Q}{d \ln \tau_{wall}} \right]$$

$$\dot{\gamma}_{wall} = \frac{Q}{\pi R_p^3} \left[3 + \frac{1}{n} \right]$$

Shear Flow Rheometry Experiment

- What are the Measured Variables?

–Flow Rate $Q = v_p \pi R_p^2$

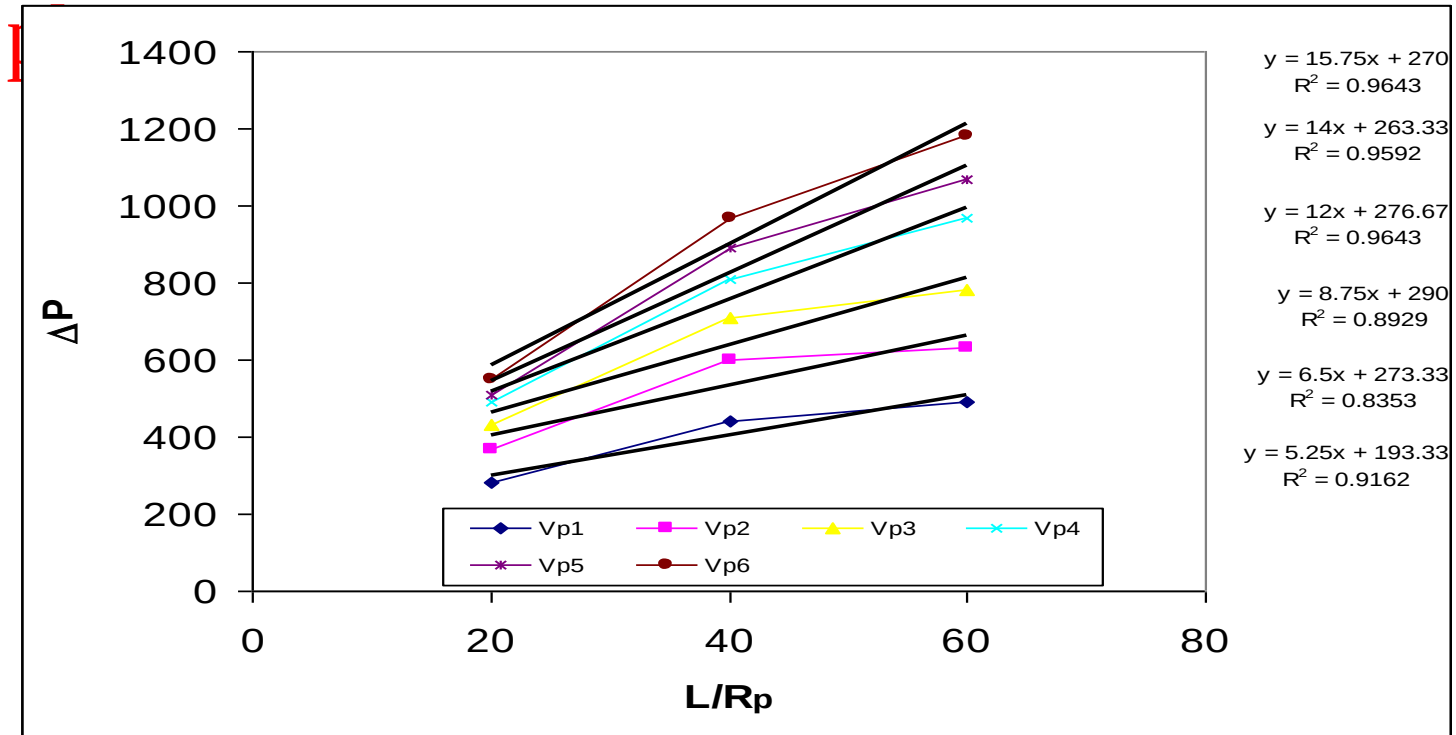
–Pressure Drop ΔP

Capillary: 30/1			Capillary: 20/1			Capillary: 10/1	
v_p (mm/s)	ΔP (psi)		v_p (mm/s)	ΔP (psi)		v_p (mm/s)	ΔP (psi)
0.01	490		0.01	440		0.01	280
0.02	630		0.02	600		0.02	370
0.03	780		0.03	710		0.03	430
0.04	970		0.04	810		0.04	490
0.05	1070		0.05	890		0.05	510
0.06	1180		0.06	970		0.06	550
0.07	1280		0.07	1020		0.07	580
0.08	1350		0.08	1070		0.08	620
0.09	1450		0.09	1110		0.09	640
1	4100		1	1170		1	670

Bagley Plot

- What do we

➤ ΔP versus L/R_c



- Information obtained?

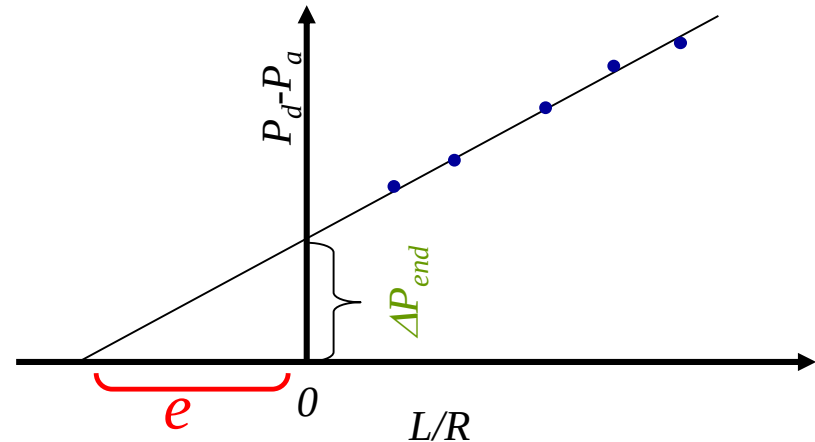
Bagley Plot

- Obtain Bagley Correction:

$$\tau_w = \frac{P_d - P_a}{2(L/R + e)}$$

- e = intercept/slope or when ΔP is zero:

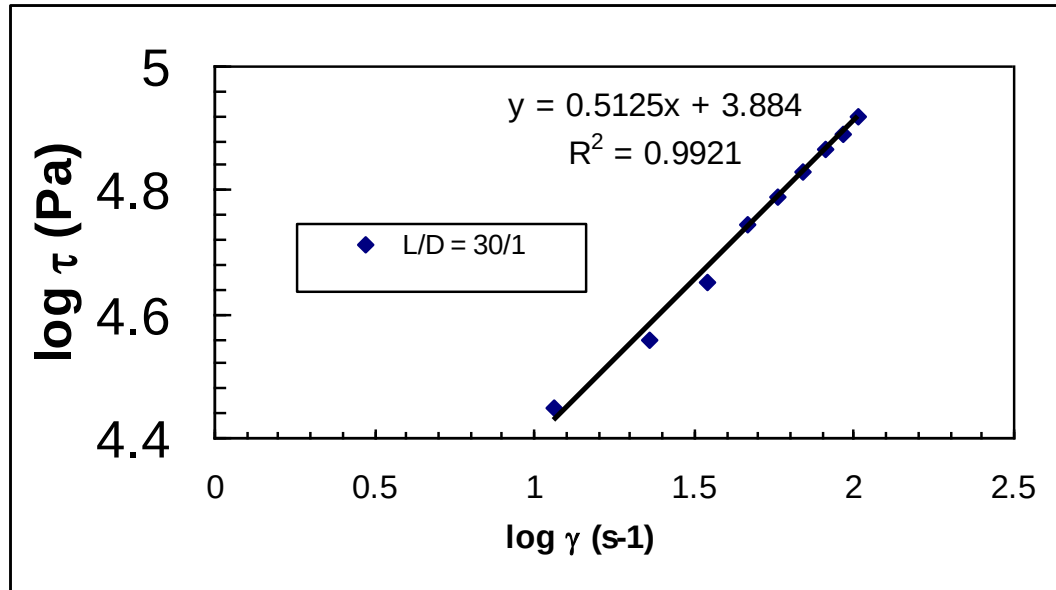
	<i>x-intercept</i>
Vp1	-17.1429
Vp2	-18.7857
Vp3	-23.0558
Vp4	33.1429
Vp5	-42.0508
Vp6	-36.7619
<Bagley factor>	



- Correct for stress τ_w with e as |<Bagley Factor>|.

Weissenberg-Rabinowitsch Correction

- Plot $\log \tau_w$ versus $\log \dot{\gamma}$.



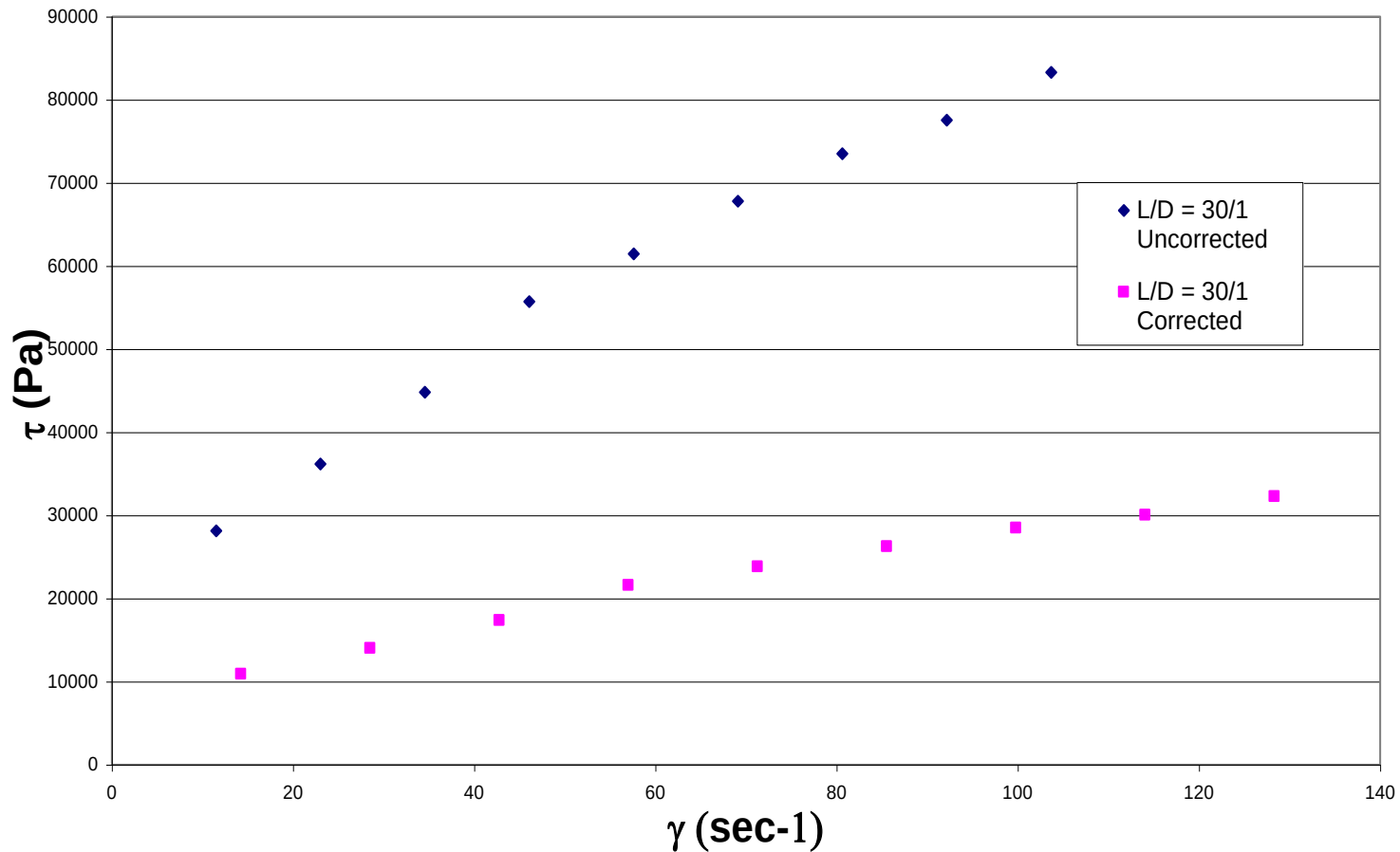
- Slope = n (for a power law model).
 - $n = 0.4256$ for 20/1 and 0.3728 for 10/1 L/D ratios

- Calculate the corrected shear rate $\dot{\gamma}$

$$\dot{\gamma}_{wall} = \frac{Q}{\pi R_p^3} \left[3 + \frac{1}{n} \right]$$

Shear Flow Rheometry Experiment

Comparison of Uncorrected data

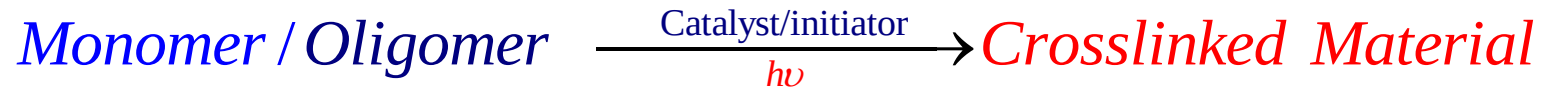


Gels and Networks

**Probing microstructure, state and kinetics
through rheology**

Gels and networks

Chemically x-linked systems



Physical Gels (reversible)

Temperature, light, hydrogen bonds

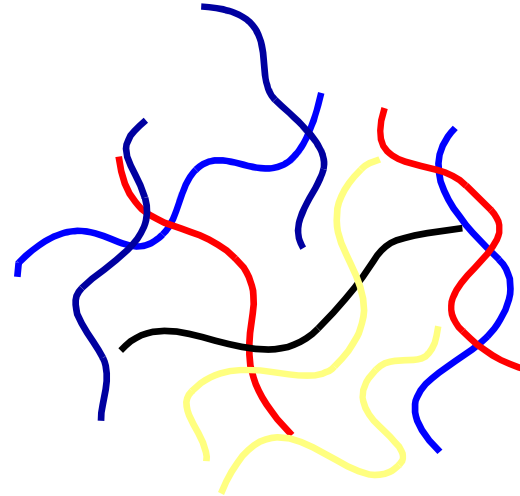
What is a Gel?

3-Dimensional network spanning the sample

Gels and networks, cont'd

Polymer chains:

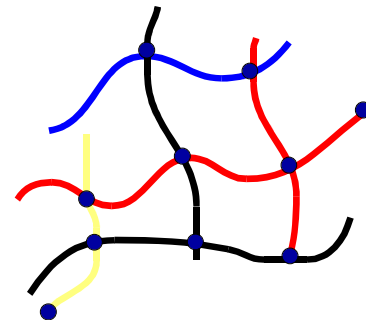
Chains can move/wiggle around through “reptation”



Gels:

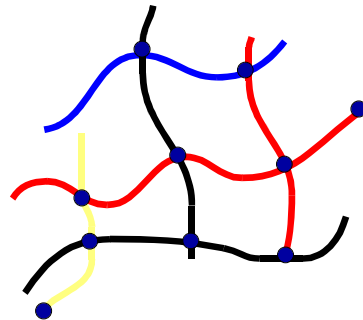
The polymer chains are constrained

Junctions can be fixed chemically or physically



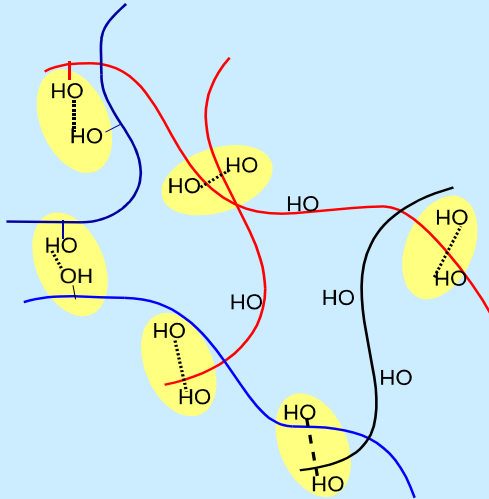
Chemically x-linked gels

Fix Junctions chemically $\Rightarrow$ Covalent bonds
 $\Rightarrow$ Chemically x-linked gels/system
(Chemically cured system/thermosets)

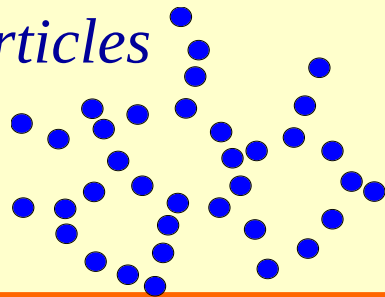


Physical Gel-Examples

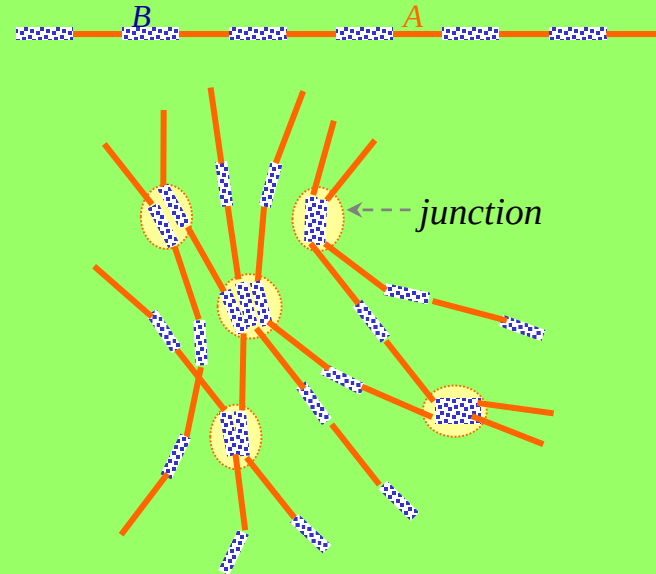
Systems with Hydrogen bonds



Colloidal Particles



Block Copolymers

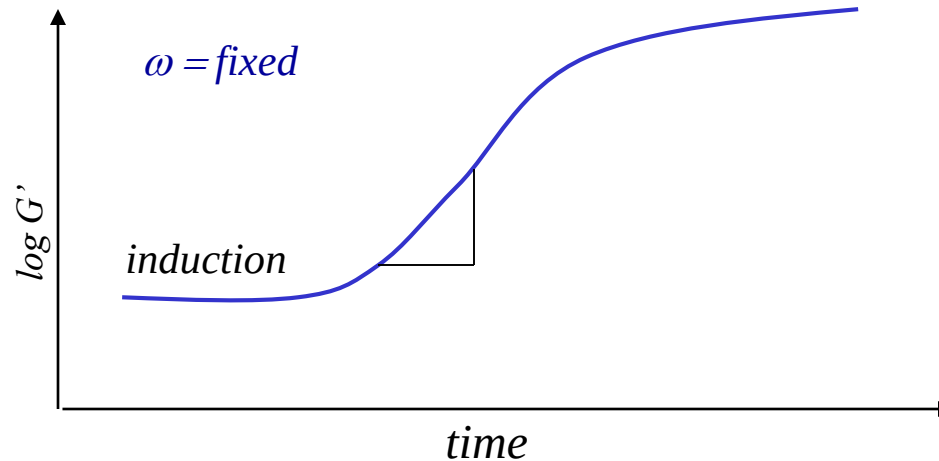


- *Ionic interactions*
- *Hydrophobic interactions*

Critical Issues

1. *What is a good way to monitor gelation/x-linking rheologically?*
2. *How can one differentiate between a gel and a sol/solution?*
3. *How can one get the gel point?*

Monitoring Gelation Kinetics

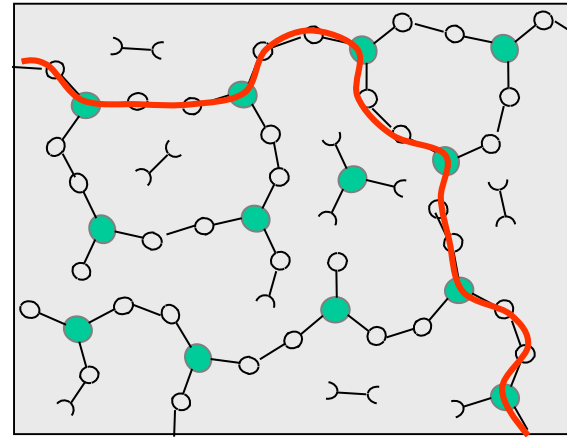
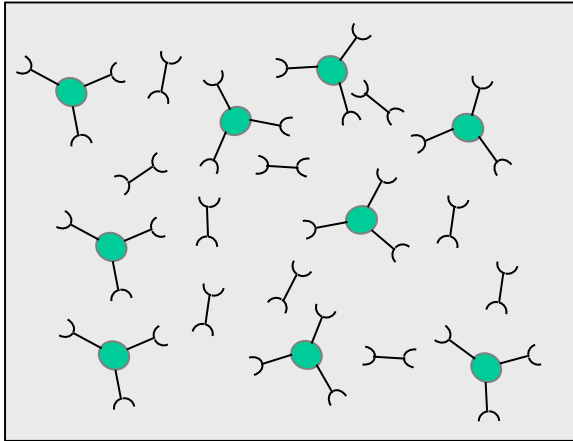


$$G'(t) = n(t) kT + G_{en}$$

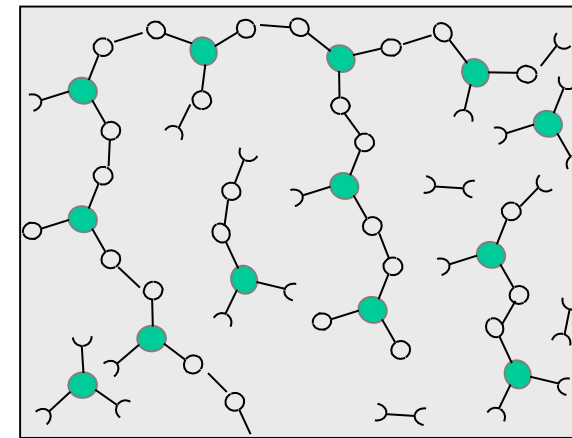
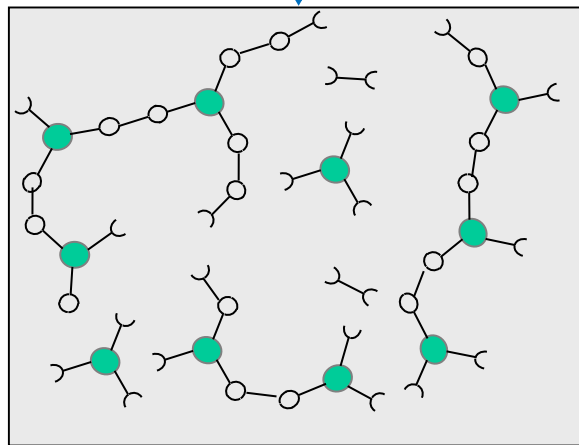
Equivalent to # x-links

$\frac{dG'}{dt}$ is used to relate to kinetics

Microstructure Evolution During X-linking

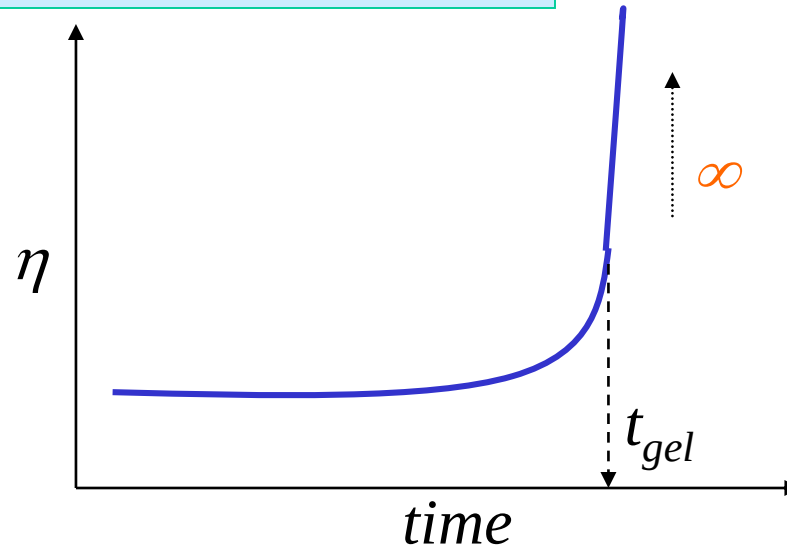


Gel point



Gel point detection – Approach 1

1. Viscosity versus time behavior

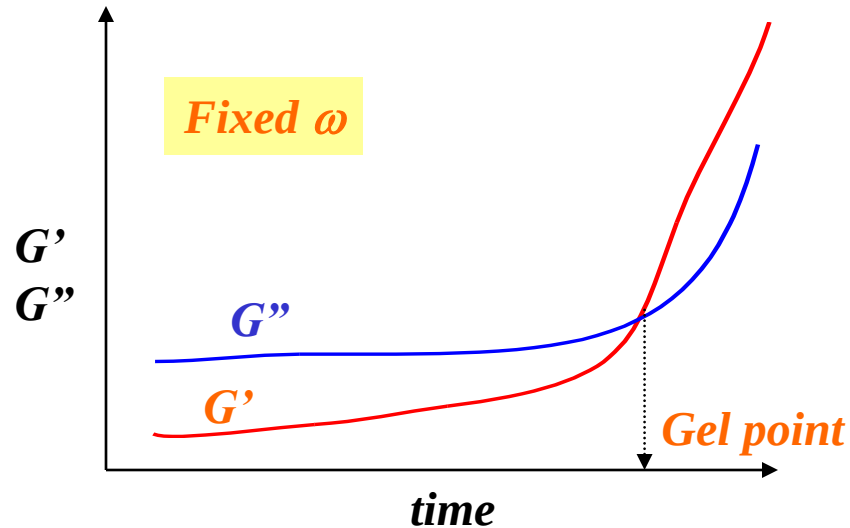


Time at which $\eta \rightarrow \infty \Rightarrow$ gelation time $\rightarrow$ Gelation process

Any Problem? shear can interfere with crosslinking/gelation

Gel point detection-Approach 2

2. G' , G'' versus time behavior



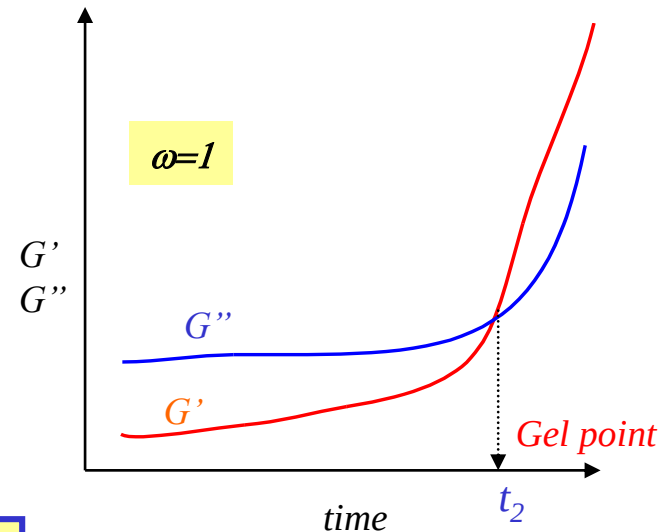
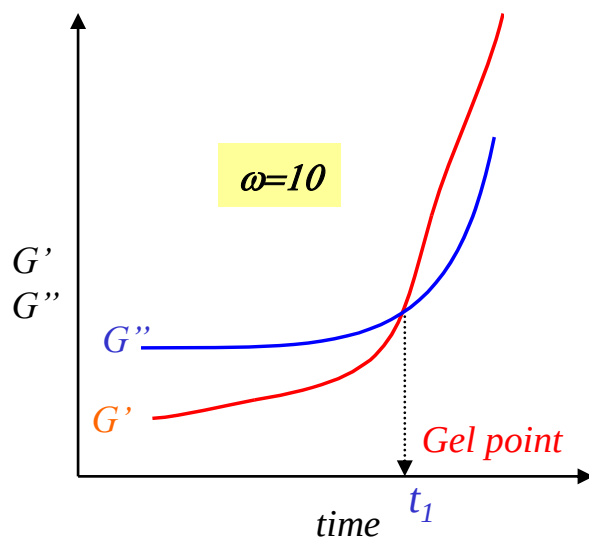
Intersection where $G'=G'' \Rightarrow$ Gel point

Problem: *Do we get same point for different frequencies?*

Gel point detection – Approach 2, cont'd

2. G' , G'' versus time behavior

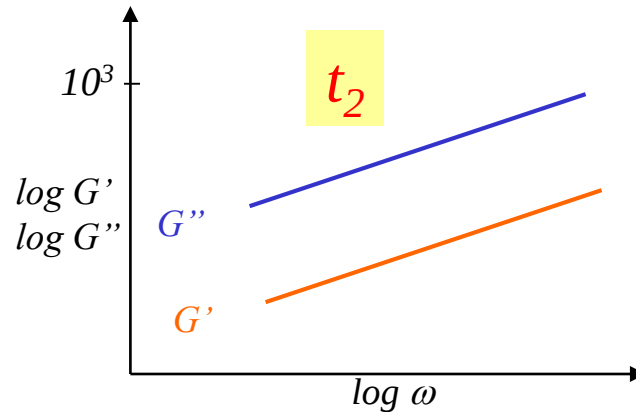
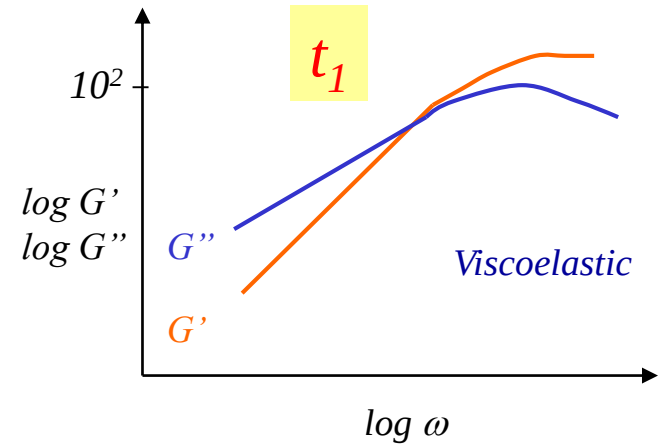
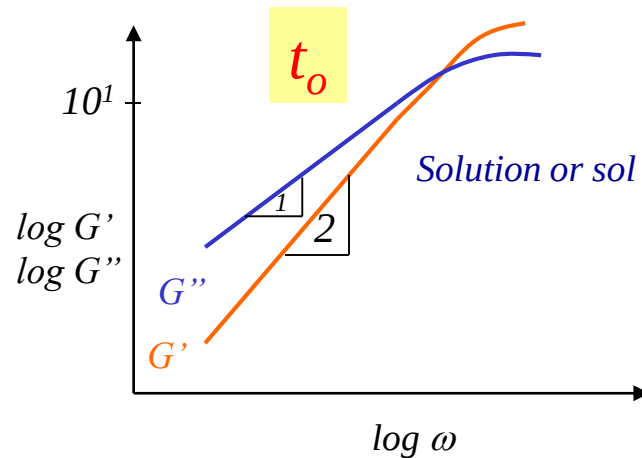
- The technique is ok if gel point is frequency independent. But often this may not be the case.



$$t_1 \neq t_2$$

Gel point detection – Approach 3

3. G' , G'' versus frequency at different time intervals



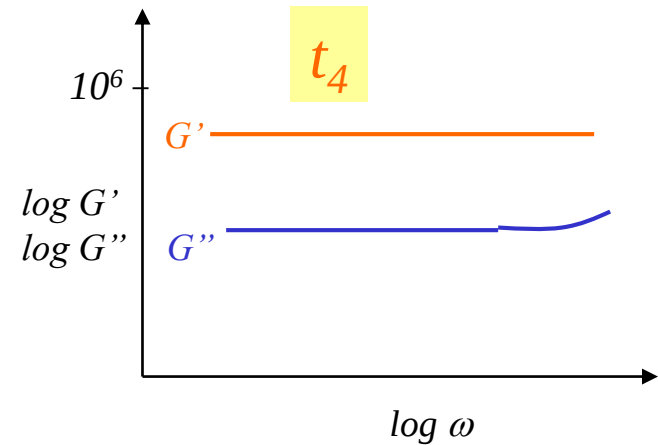
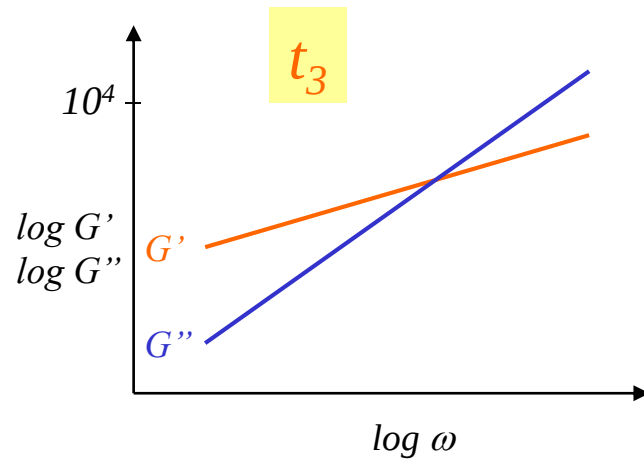
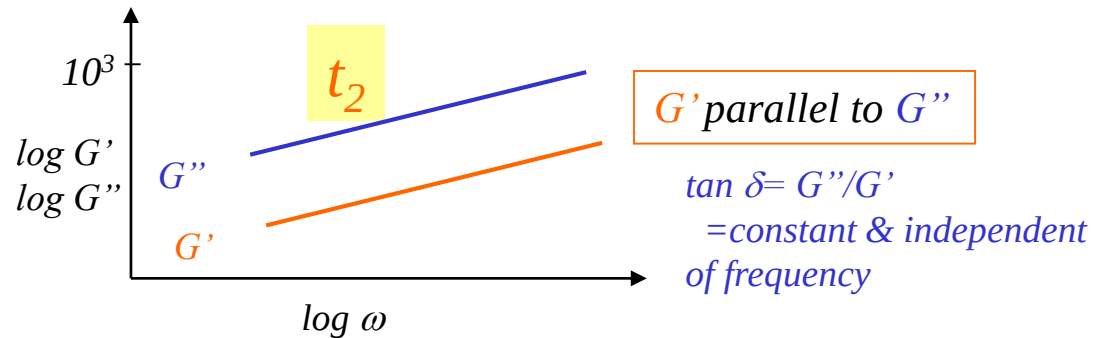
G' parallel to G''

Gel Point

Winter-Chambon
criteria

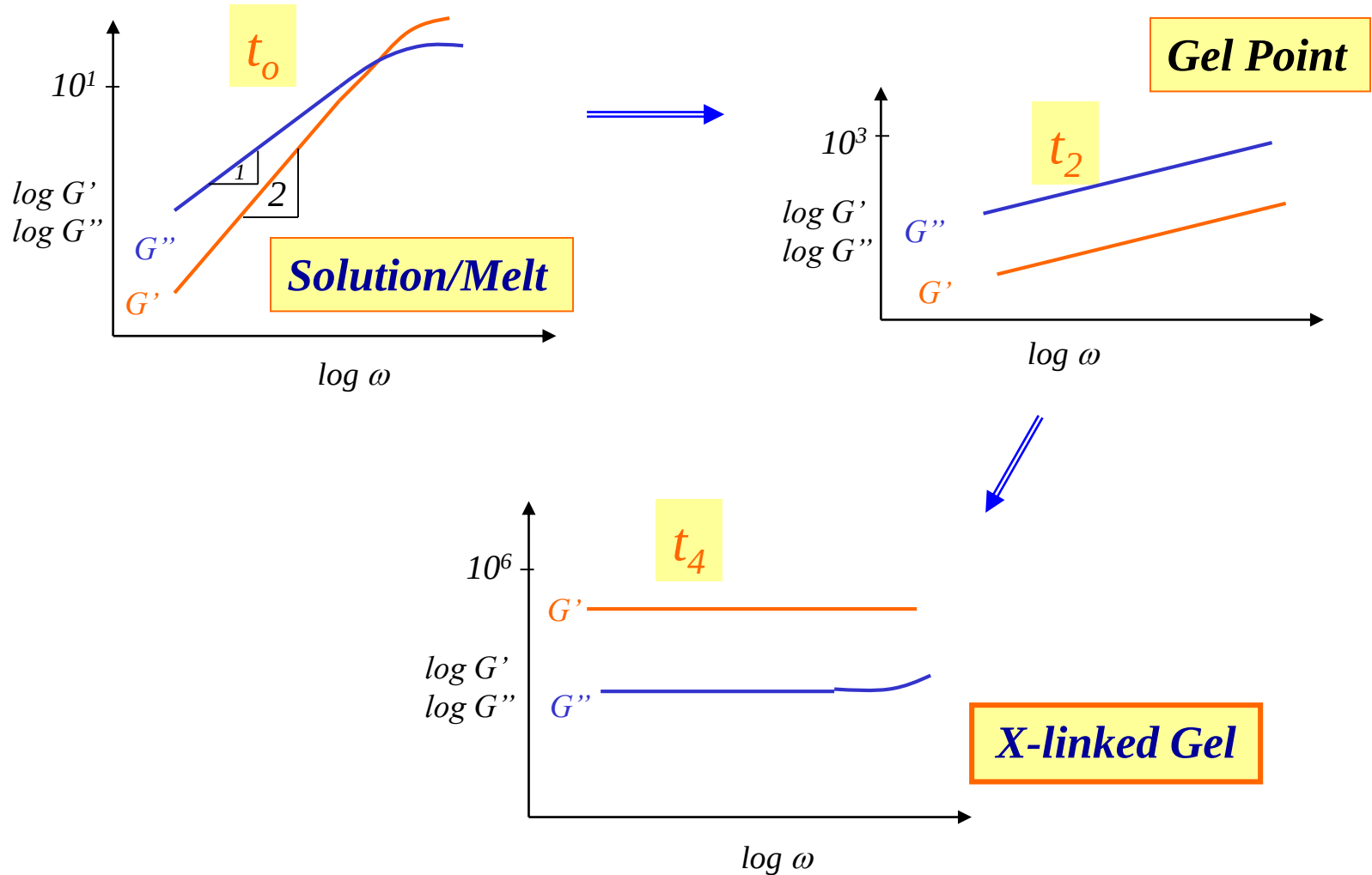
Gelation & Gel point detection

Gel Point



$G' \gg G''$
 G', G'' frequency independent
 = **Fully x-linked gel**

Recap- Gelation & Gel point detection

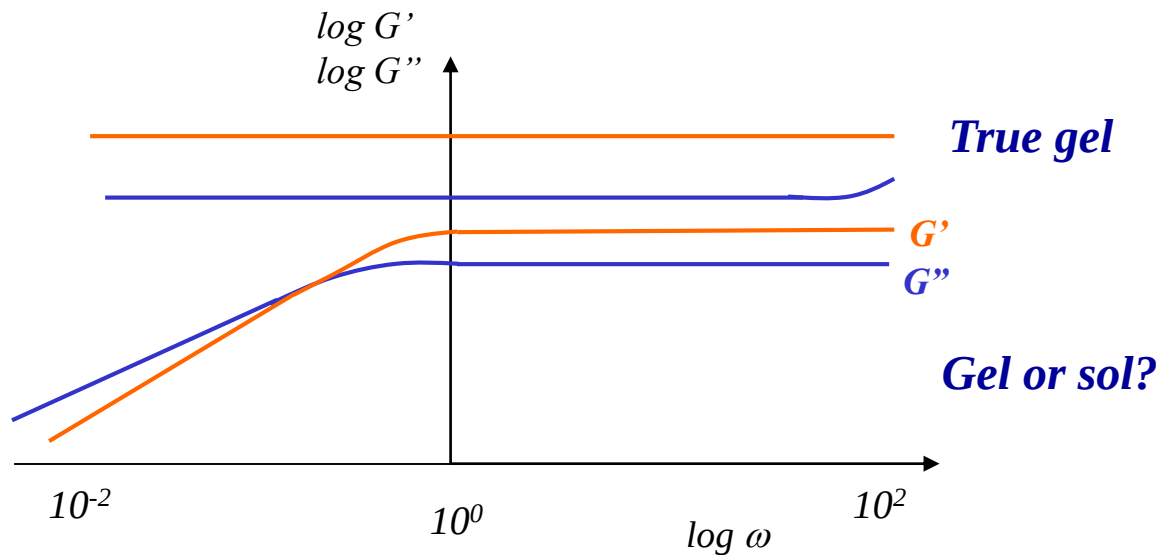


At the Gel Point

- *Molecular weight M_w goes to infinite*
- *System loses solubility*
- *Zero shear viscosity goes to infinite*
- *Equilibrium Modulus is zero and starts to rise to a finite number beyond the gel point*

Exercise – Is this a gel or solution?

Go to lower frequencies to insure G' and G'' are frequency independent



Back to Gel point analysis

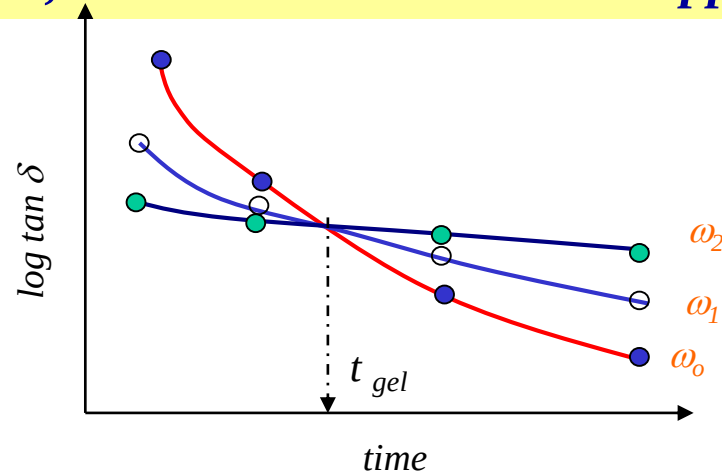
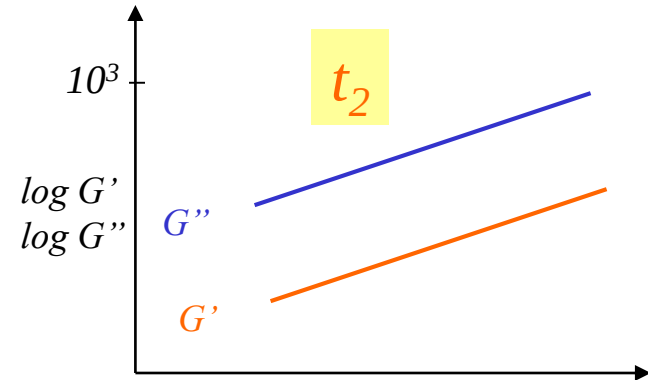
Concern:

*What if we have large enough time interval
so we miss gelation point (t_2 behavior)*

Solution:

Plot $\tan \delta$ versus time

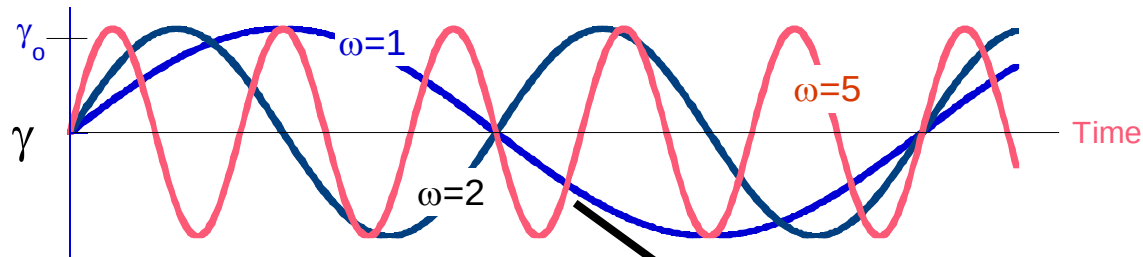
For fast reactions, in addition use Multiwave approach



Multiwave Approach

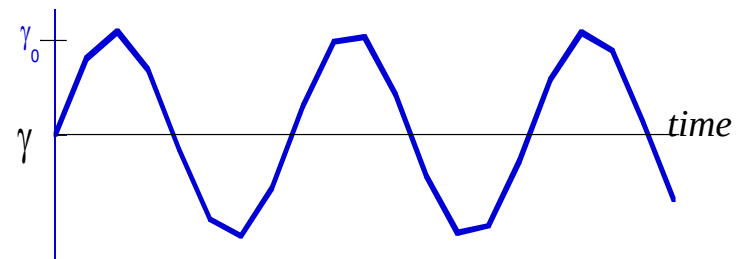
Combine several frequencies/oscillation into one.

Fourier Transform Mechanical Spectroscopy



Input:

*Strain wave combination
of several sine waves*

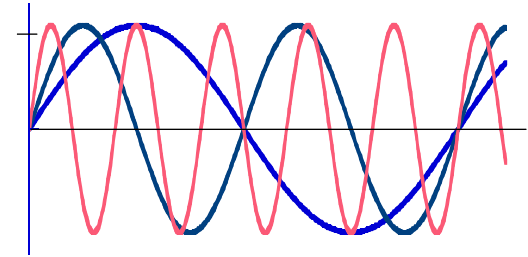


Output:

Complex torque signal $\xrightarrow{\text{Fourier transform}}$

G', G'' for several frequencies from one data point

Multiwave Approach, cont'd



Advantages:

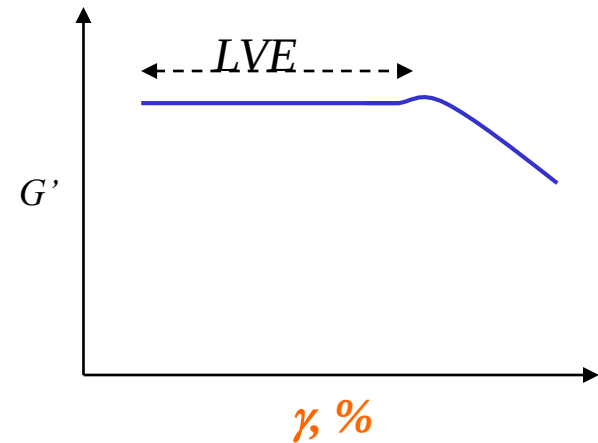
- *Effective in fast reacting system*
- *Only one data point gives several frequencies*
 - Can get G' , G'' versus ω from 1 data point*
- *Can monitor x-linking quickly*

Multiwave Approach, cont'd

Pick frequency ω_1 : other frequencies have to be multiples of ω_1

Input is superposition of multiple oscillatory strains: $\gamma = \sum_{i=1}^m \gamma_i \sin \omega_i t$

Linearity condition
 $\gamma_1 + \gamma_2 + \gamma_3 + \dots + \gamma_{10} < LVE \text{ limit}$



Next Class

More on fractals – what is it?

Real data – gelation and gel point

Physical vs chemical gels

Thixotropy